

Bacterial

Food Poisoning

Bacterial food poisoning results from the consumption of contaminated food. Food poisoning bacteria can be classified into two

1. Those that infect the body e.g. *Salmonella* group
2. Those that produce toxins during their growth in the food. They include *Staphylococcus aureus*, *Clostridium perfringens* group A and *Clostridium botulinum*.

To diagnose food poisoning, it is necessary to know the symptoms and time of onset.

Salmonella Food Poisoning:

Source: Man, animals, birds. The organism is killed by boiling, so contamination is post-cooking. Infection occurs during the process of eating from improperly cooked food. Common in foods such as meat products, cream and eggs.

Incubation: 12-24 hrs

Pathogenesis: Organisms multiply in the small and large intestine and cause epithelial cell damage. Diarrhoea & Pyrexia are common.

Specimen: Faeces and the contaminated food. The food should be taken to the laboratory in its original container or transferred aseptically to sterile containers. Refrigerate the food until processed.

Clostridium botulinum Food poisoning

Source: Soil & animals, Contamination is pre-cooking, common in canned and home preserved food such as meat, vegetable, pickles.

Incubation: 12-72 hrs

Pathogenesis: Organisms multiply in food and produce heat-labile potent neurotoxins. Food poisoning may result in respiratory paralysis and death. Prognosis is severe.

Specimen: Suspected food, patient's faeces and vomit specimens

Staphylococcus Food poisoning

Source: Nose & skin of man. The organism is killed by heat, so contamination is post cooking. Infection occurs during eating of the food. The organism produces enterotoxin. The infection is common to foods such as cream, meat, pork etc.

Specimen: Faeces of the patient is to the laboratory for examination. It may be necessary to take a nasal swab by the catering staff for screening.

Clostridium Perfringens Food poisoning:

Source: Man and animals. Contamination is pre-cooking especially of meat cooked in bulk. Spores of the organism survive heating. Upon cooking, the spores germinate and cause diarrhea when ingested in food. The organism produces enterotoxin in the gut.

Specimen: The food remaining and the patient's faeces are sent to the laboratory for culture.

Bacillus cereus: It is a saprophyte that contaminated heat treated milk or boiled rice. Food poisoning sets in after consuming the food especially boiled rice that has been stored for long periods after cooking. The organism is isolated, identified and typed.

Specimen: Patient's faeces

Many other organisms are capable of causing food poisoning. Infected food usually makes, not show any change in taste, smell, or colour.

Helicobacter pylori

Viruses - that can cause food poisoning

- 1) Norovirus
- 2) Hepatitis A virus
- 3) Rotavirus
- 4) Astrovirus
- 5) Adenovirus type 40 & 41

The viruses are transmitted through contaminated food, water, or surfaces that can lead to symptoms like nausea, vomiting, diarrhea and abdominal pain.